What About Tests In Smart Environments? On Possible Problems With Common Sense In Ambient Intelligence

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Abstract

Ambient Intelligence seems to be a perfect field to evaluate common sense reasoning algorithms as environments full of smart transmitters and receivers could resemble the real world where human senses the surroundings and processes input to use it in an intelligent way. But is it real enough? Will it be really easier to use common sense processing to predict what somebody's behavior means? By simple experiments with automatically created pseudo-Schankian Scripts we will show that naturalness of human action varies strongly - probably depending on evaluator's personal background, experiences and imagination. To limit the contextual situation we created virtual house by describing it with words for actors, places, tools, objects and motions common for such a place. Then we made the system check the commonness of actions randomly performed in this environment and see if an action is significantly against common sense or not. We will show that evaluation of naturalness becomes clearer and fairer when context is set but we will also call into question that ambient technologies can describe the context well while they could do just opposite - they could limit it. In this paper we also describe other problems of common sense when used in AmI and suggest some possible solutions.

1 Introduction

It has been almost a half of century since Mortimer Taube wrote the first anti-AI book, "Computers and Common Sense: The Myth of Thinking Machines"[Taube, 1961]. Although it did not receive much attention, it clearly shows that from the very beginning of less or more advanced technologies people doubted that machines could process the vastness of knowledge called "Common Sense" (in this paper often abbreviated to CS) which we gather from the first moments of our lives. It has been always called "the Holy Grail of AI" but none of the developed systems could be claimed fully successful. The turn of the century brought us repositories as CyC being built by specialists, OpenMind CommonSense being created by mass collaboration of Web users[Singh, 2002] or ThoughtTreasure built mostly by one person[Mueler, 1998] and lexical databases like WordNet[Fellbaum, 1998]. Gathering common sense knowledge entries became main trend in the field of CS and we also concentrated on this task, though we try to achieve more automatic acquisition methods. However, the more experiments we performed, the more they made us doubtful about the main-stream "collecting approach". The main problem - lack of contextual flexibility which brings the lack of universality - is still being ignored. We can collect plenty of commonsensical data about some objects and their properties but the usability of this knowledge will change in different situations or with different actors. Is Common Sense really universal knowledge? It surely changes its commonness depending on the situation and we showed it in a very simple experiment. Evaluating such knowledge without context does not make much sense but is it the Ambient Intelligence where Common Sense researchers could prove their systems work well? When Locke[Locke, 1690] proposed the term "common sense" he meant a sense which combines input form other senses into "human understanding". At first sight Ambient Intelligence seems to give us what common sense researches lack most often nowadays - real applications with real input from real environment. From the arificial senses which could create Lockian Common Sense. However, a kitchen stuffed with sensors, RFID tags / readers and intelligent fridges still does not understand what is really going on inside it except predicted input combinations. Even if everyone wears "smart clothes" and the system gets hold of who does what - it will not necessarily be able to recognize what such behavior pattern could possibly mean and to what consequences it could lead. The problem of preprogramming every possible scenario will remain and Common Sense in Ambient Intelligence will be common only for a given ambient even if sharing data from similar devices of other users. In the next section we describe above mention experiment demonstrating problems of common sense evaluation and then we will present a model scenario in a fashion of famous ISTAG's scenarios for 2010[Ducatel et al., 2001] explaining problems and our ideas for solving them.

2 Confirming Some Obvious Things About Common Sense

Before we go to the crux of the matter, we will briefly describe the experiment which confirms facts most of us feel to be obvious but which are widely ignored by "Knowledge Collection-Oriented" main-stream of current CS research.

2.1 Automatic Minimal Script Retrieval

In our research we use several languages from different language groups but here we will introduce results of experiments performed with Japanese (to implement our method to some language it is enough to change the grammar forms for retrievals, for example to use English prepositions instead of Japanese particles). The system analyses one pattern sentences describing simple actions in form of one noun, one particle and one verb and tries to retrieve most common previous and following actions imitating an idea of Schankian scripts[Schank et al., 1998]. A sentence (Place)-Noun (+place particle) Noun (+object particle) Verb (called here "motion") is inputted and such combination is being confronted with the Web. Then the neighbors are retrieved for extracting previous and next actions (for faster retrieving we use web-based bigram and trigram dictionaries of nouns, particles and verb combinations). This algorithm allows to automatically create minimal scripts, for example "Give birth to a child → take care of (it) → do best" or "wait for friends → have fun → go back home". The next step was to evaluate these retrievals.

2.2 Experiments and Results

We picked results of 34 inputs made by users. Wide range of different words was chosen as we wanted to keep the broadness of common sense knowledge - any kind of action was allowed and only thing we set was telling users that these action would be faced and analyzed by house-helping devices. We presented 10 judges an evaluation list where every input was also divided into "place + motion", "object + motion". Additionally possible previous and following actions were proposed. Every such output was also labeled by the system as "natural" or "unnatural" and a user was to mark such label in a scale from 0 to 5 in two levels: how precise the label is and how useful it would be for an intelligent house. Previous and next actions were evaluated for naturalness instead of preciseness. The average marks given by ten judges for the preciseness of automatic naturalness labeling were 66.4% for the whole action sentence, 85% for object-verb form and (quite surprisingly) only 57% for "place+verb" form. The usefulness for these three forms was evaluated similarly : 61.2%, 80.2% and 54%. Previous actions were evaluated natural in 67.8% and useful in 61.2% cases while next actions were evaluated natural in 55% and useful in only 45.5% cases. After taking a closer look to the evaluation data, it became clear that what is *natural* and *useful* depends deeply on individual factors which are hard to estimate - probably from own experiences to richer or poorer imagination. Figure 1 shows an example of opinion dispersion among judges.

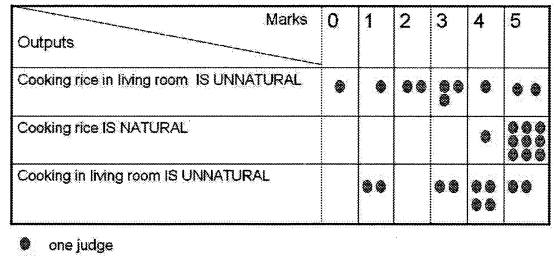


Figure 1: Example of opinion dispersion while evaluating system's measurement of common action naturalness

3 When "Cooking Rice" Becomes Uncommon?

It can be said that thinking too much about commonsensical happenings distorts the commonness of these happenings. Deep thinking about shallow things may be one of the key problems in this case. As seen in the example in Figure 1, *cooking rice* is of course natural for every judge (output was judged by native Japanese) but ideas about naturalness of cooking and rice varies heavily depending on the place. Very wide expressions are easy to be agreed on, however more details triggers some psychological mechanisms and the ideas of commonness start to vary. To keep everyone's imagination focused and still be able to increase the number of details, we decided to limit the user's imagination by preparing more detailed experiment environment.

3.1 Limiting User's Imagination

As we are thinking about implementing commonsensical knowledge into a intelligent system helping at home, we decided to set a simulation environment at the house with limited number of actors (4, e.g. child, dog) performing limited "motions" (24, e.g. to eat, to open, to bring) with limited objects (32, e.g. knife, pot, trash) in limited places (8, e.g. toilet, balcony, shelf).

3.2 Experiments in More Specific Ambient

Random actions were created in following form: actor (subject particle) - place (place particle) - object (object particle) - verb ("motion"). When motions as "giving" were involved, "place" was being replaced with another "actor" chosen randomly from the set. Then the system was checking the Web frequency for every combination of actor, place, object and motion. Depending on these results, the system was 1) approving the action as natural (15% of cases) 2) suggesting which part is unnatural (55%) or 3) claiming that action has too many uncommon parts that is not worthy analyzing (30%). For instance "mum gives a child spaghetti" was evaluated natural, "a kid puts back a soap to the balcony" drew suggestion that "soap" is unnatural part of this action and "a robot opens a rice-cooker at the toilet" was rejected as being "too strange". Then we made 14 judges familiar with the environment and actors evaluate 40 outputs. The average evaluation estimated in the middle as in the experiment described in the previous section but what we were interested

in was the opinion dispersion. In comparison to the previous experiment, the number of asymmetric or flat evaluations (where judges opinions varied heavily) dropped from 57% (only the whole actions were considered) to 15%. Most of the dispersed opinions was on actions where robot was an actor or balcony was a place.

3.3 Conclusions From This Trial

Three conclusions may be drawn:

- Very obvious truths are evaluated natural and useful but the context may make them unclear, less useful or even untrue;
- Common sense knowledge should be gathered together with context or, as is our choice, should be retrieved on spot for given context;
- Technologies allowing to recognize given situation of real life should be a good place to test CS algorithms.

As it is easy to guess, the last conclusion led us to Ambient Intelligence. Common sense knowledge system experiments should be performed in environments where programs are tested in the same situations with the same set of details. Although we specialize in retrieving common sense knowledge from the Internet, we felt that Ambient Intelligence could become very convenient training ground. Surely a discussion on how to stimulate progress in interdisciplinary field of CS reasoning should be done together with AmI specialists, hoewver the open question is - will AmI lead to achieving Holy Grail of AI? In our opinion it will help but we have to be really careful. We need as much sensor input as possible but we must remember how specific is the aim of AmI. Gathering knowledge about one user, one group or one environment will tend to build semantic dependencies concentrating on market-driven choices of input. The holistic approach for learning CS from all sensors data will be simply impossible just because companies and users will naturally protect their data. One of the possible solutions could be sensor input simulation which basic ideas were already introduced back in 2003[Rzepka et al., 2003]. Before explaining our ideas on this matter, we will present a short scenario showing some of the possible problems.

4 Scenario 5: 'Susan' - Housewife

After not so long but sound sleep[1], Susan wakes up at her bedroom. Her Private DayScheduler (PDS) had no reason to wake her up as it didn't recognize anything in "TV-guide" which could be interesting for her. If it did, the program would be recorded and the day rescheduling would be performed depending on the tasks planned for this particular day. Her kids have gone to school already after being waken up by their PDS - they are big enough to eat what Smart Kitchen suggested Susan yesterday. She cares much about what her kids eat everyday and Smart Kitchen always helps her to choose new healthy products which people recommend in their blogs. But today Susan has one problem which is her terrible headache. The Smart Kitchen discovered her unusual state but had no idea that is because of drinking too much gin and tonic on the previous night. Susan asks for a pain-killer and it soon helps with her headache, however she feels heavy raising herself up and taking some milk from the fridge. Her movement caused the preprogrammed reaction - the coffee maker switched on. For Susan the scent of coffee during the hangover is the worst thing she can imagine, so she feels bad again and goes out to take a deeper breath of fresh air. Her PDS reacts immediately assuming that the shopping planned for the afternoon is now rescheduled. Within a few seconds she hears the beep and a little screen recommends her the nearest grocery. A bar code for all the things which must be bought appears - now it is enough to bring it to the store where special assistant will suggest the recommended equivalents. It (assistant is a little tool) already knows that Susan loves novelties even if the price is a bit higher than usual. But she is not in mood for shopping. She enters the park and sits on the bench dying for some rest. PDS beeps constantly trying to show her way to the grocery, so she switches it off. She knows that the security system calls her husband and the nearest police station if her PDS goes off in a dangerous place. What she does not know is that this park has been just listed on the Metropolian List of Dangerous Places. Somebody had his intelligent clothes stolen here last night...

[1]Similarity to the ISTAG scenarios is intentional however the purpose of writing it is a bit different.

5 Sensors Need Help As Well As We Do

The simplest way to avoid some of the situations described above would be a natural language interface which asks what is going on with Susan and understands the possible consequences of having hangover. However, people are not always in mood to explain a machine their feelings. It would be much easier to guess what is happening if commonsensical knowledge as "too much gin and tonic in the evening = possible hangover in the morning" was taken into consideration. Of course questions as "how much is too much" or "what does it mean in numbers that human becomes slower during the hangover" remain unanswered until Susan repeats her "drinking" and "feeling unwell" actions several times. But what does it mean that she drinks every night? Why her busy husband is not informed about it but he has a call about her resting on the park bench? Analysing human behavior just upon partial input will always be dangerous but the more possible conclusions are drawn, the less likely it is to be sure about the situation. The machine needs other sources of confirmation to recognize danger correctly or to avoid turning user's life into a nightmare where he or she is afraid to do anything unusual just to avoid possible problems with gadgets designed to make their life easy. Socio-political factors are not the only issue here, we need solutions already on a very prosaic level. In next subsections we introduce some of our thoughts which could bring some help or at least pass some hints to AmI developers.

5.1 Digging Treasures In the Garbage Can - Common Sense From the Web

As we already mentioned above, we retrieve common behavior patterns from the WWW. This is one of the practical

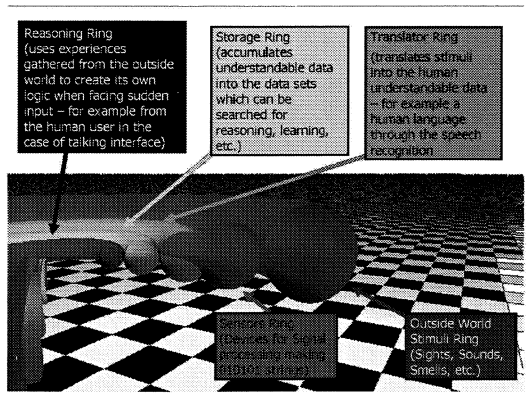


Figure 2: Representation of A.I. in a shape of five rings showing flat dependency between outside layer of real-world and inside layer of processing outside data to e.g. learn commonsense knowledge.

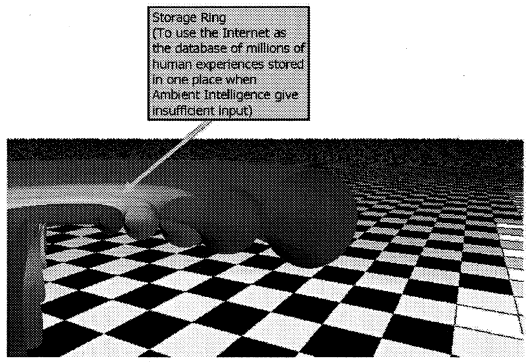


Figure 3: Our idea of supplementing the missing data by text mining the Web in order to achieve missing or ignored data.

examples of our philosophy which could be represented as shown in Figures 2 and 3.

The main assumption is based on observations nicely described in James Surowiecki's bestseller "The Wisdom of Crowds"[Surowiecki, 2004] and says that the ordinary people's knowledge is almost always correct especially if the specialist opinions are mixed with jokes or very silly opinions. Receiving a global input from all the devices of AmI could help us a lot - but as we mentioned before - the business would not let it happen and even if open source version would arise - the users would probably avoid giving away their private data. AmI will need the support of the Web - if not semantically, then at least as a model to mine behaviors just as web-mining analyses browsing patterns now. However, our claim is to retrieve lexical structure in order to build semantic structures to cooperate with any input from the outside world as any of such input can be represented in natural language. It is the point when our approach of "Crowds-supervised learning" starts. The next step, or rather the step to be made simultaneously, is an universal theory (or rather assumption) for all humans behaviors.

6 AmI And Global Ethics

We presented our idea of Crowds being ethically correct during the first Machine Ethics Session of AAAI Fall Symposium in 2005[Rzepka, 2005] although we were developing algorithms telling Good from Bad years ago[Rzepka, 2002]. Last year we presented a supplement proposing to calculate Bentham's Felicific Calculus[2] by using our techniques[Rzepka, 2006]. If we succeed, the program should be able to measure following variables (vectors):

- Intensity - how strong is user's feeling? "Losing a parent" and "losing a watch" is a different world but it is hard to conclude from the lexical surface only - affective computing must be performed in order to measure the intensity of possible human reactions.
- Duration - how long an emotion lasts? What is the difference between "one hour party" and "whole night party"? Will it be better for the user if I set up one minute of his favorite music or one hour?
- Certainty or Uncertainty - how sure can I be about being punished or awarded? Buying will assure human the pleasure but counting on somebody giving me something leaves the risk that I will not feel the expected amount of pleasure because I will not get what user wanted.
- Propinquity or Remoteness - how close is the positive/negative emotion from the user? What is the difference if his close friends has won something than a case where such thing happens to my neighbor.
- Fecundity - the probability that one sensation will be followed by the same kind of sensation. Will a good state of something remain good or is it going to worsen? Will my decision keep my user in good mood?
- Purity - the probability that one sensation of the opposite kind will be followed by sensations of opposite kind. By default the machine should always try to preserve user's good emotional state and do everything to avoid worsening it.
- Extent - the number of users to whom my decision or some happening extends. For instance if the machine's decision affects one coworker of its user instead of ten, it is preferable choice.

Algorithms based on utilitarian ideas will probably have more acceptance within AmI communities than in other fields but still there will be a large margin for discussion on hedonistic

[2]Jeremy Bentham's famous quote "Nature has placed mankind under the governance of two sovereign masters, pain, and pleasure. It is for them alone to point out what we ought to do, as well as to determine what we shall do?"[Bentham, 1789] is controversial, we agree, but probably it is easier to implement than Kantian ethics and we will avoid the discussion which philosophy fi ts us most!

background to all human behaviors. But on the other hand using evaluation techniques on common happenings and objects could lead to safety valve letting the system recognize danger even if given case was not inputted beforehand. Research on retrieving opinions about e.g. products by using specific language features could help on retrieving the emotional load of human behaviors and their consequences.

7 Final Conclusions

By setting different environments giving different set of inputs (RFID or any other kind of sensors) we would achieve a method for equal evaluation for Common Sense Reasoning specialists, their databases and algorithms. In our opinion we are still far away from universality of intelligent systems using Common Sense because its researchers have problems with comparing their systems. Collected knowledge should be tried not only in single, still limited applications but in constantly changing real world simulations. However, even if bumping into reality seems to be a great method for testing CS systems, it does not mean that Ambient Intelligence will solve problems of digitalizing Common Sense. In this paper we tried to show remaining problems by creating an example scenario and suggested several ideas which might be helpful for AmI. These ideas could be described in two bigger groups - "Web Support for Knowledge from Sensors" and "Crowds Ethics for Intelligence Systems". Our future work is to continue developing our methods testing them in smart environments whenever it is possible. We are also open for any kind of cooperation as Japan has implemented several AmI technologies already but our knowledge on the field is insufficient to prepare professional tests.

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